

HIGHER TIER ONLY QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Mg + 2 HNO₃ → Mg(NO₃)₂ + H₂ award (1) for formula of salt award (1) for formula of hydrogen gas and correct balancing only if salt formula is correct		2		2		
		(ii)		award (2) for all six points plotted correctly (±½ square) award (1) for any four correct points award (1) for smooth curve		2	1	3	3	3

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		credit any three of the following four award (1) for any of following • rate decreases • reaction gets slower • rate is greatest at the start • reaction is fastest at the start award (1) for any of following • reactant particles are used up (over time) • concentration of reactants decreases (over time) • there are fewer reactant particles (over time) • there are more reactant particles at the start award (1) for any of following • there are fewer (successful) collisions per second • the frequency of (successful) collisions is lower • there is less chance of (successful) collisions • less hydrogen is produced award (1) for any of following • line levels out when reaction has finished / all the particles have reacted • particles (of metal / acid) have all been used up at 25s • reaction has finished at 25s / 35cm³	3			3		3
	(b)			(catalysts) increase the rate of a reaction / speed up a reaction / make a reaction faster (1) by lowering the activation energy / by lowering the energy needed for successful collisions (1)	2			2		2
				Question 4 total	5	4	1	10	3	8